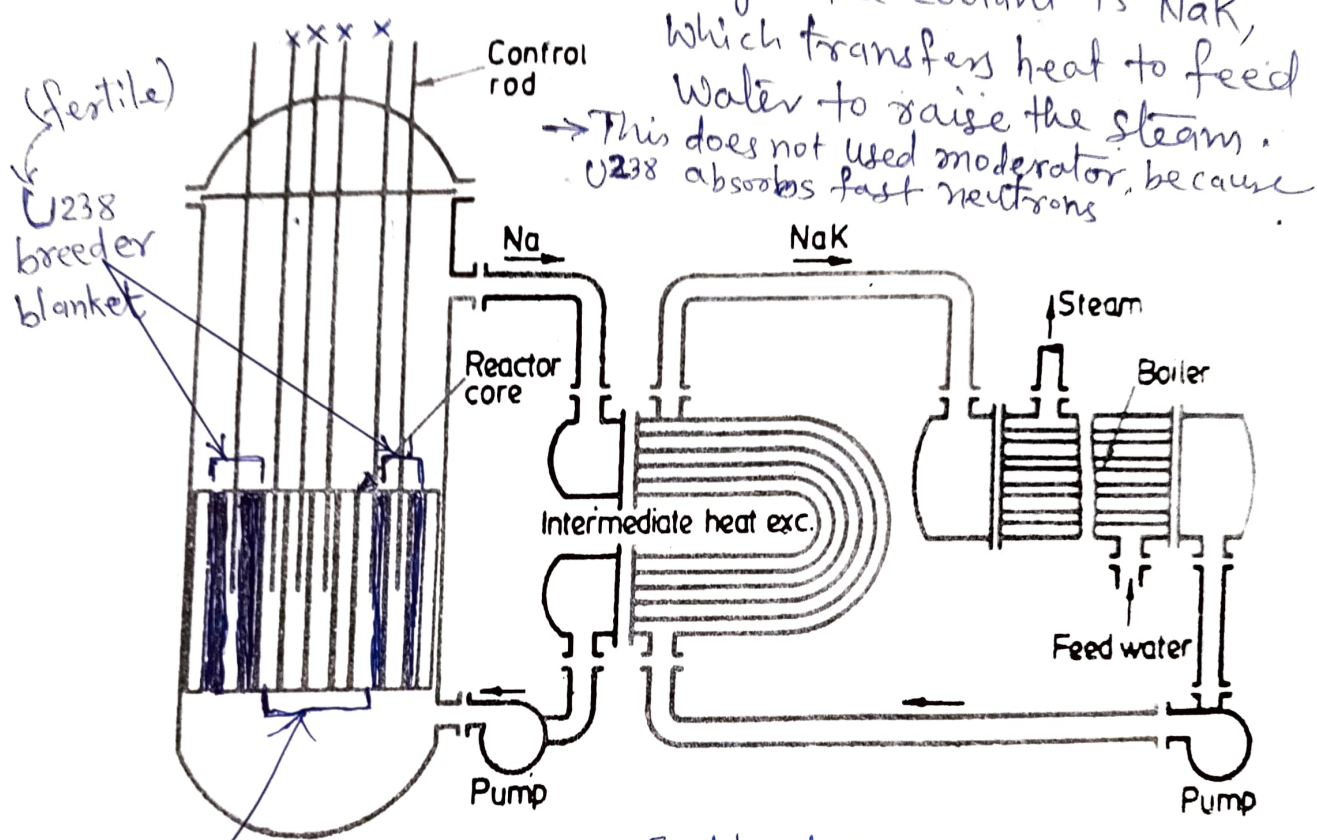


Fast Breeder Reactor: It produces heat and at the

Same time converts fertile material into fissile material.

In this reactor the core containing U^{235} is surrounded by a blanket of fertile material i.e. U^{238} which gets converted into Pu^{239} (fissile material). Two heat exchangers are used. The reactor core is cooled by liquid sodium (Na). In the secondary heat exchanger the coolant is NaK, which transfers heat to feed water to raise the steam.



Fast breeder
Fig. 5.3. Sodium-graphite reactor.

U^{235} core
(Fissile)

Breeding ratio > 1

Liquid sodium (Na) can reach high Temperature at a moderate pressure of only 7 kg/cm^2 so, Intermediate heat exchanger is necessary between reactor and the boiler. Coolant leaves reactor at 640°C and steam temperature is 540°C .

FBR also uses $20\% \text{ PuO}_2 + 8\% \text{ UO}_2$ as fuel and Thorium as ~~fast~~ fertile material which converted into U^{233} which is fissionable. [Worlds largest deposits of thorium about 4,50,000 tones in the form of sand found in Kerala and Gopalpur coast of Orissa.]

Typical power densities (MW/m^3) in fission reactor cores are

Gas cooled gas reactor	: 0.53 MW/m^3
High Temp. gas cooled "	: 7.75 MW/m^3
Heavy water "	: 18.0 MW/m^3
Boiling water "	: 29.0 " "
Pressurised water "	: 54.75 " "
Fast Breeder reactor	: 760 MW/m^3
